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Effect of stocking density on laying performance, egg quality and blood parameters of Hy-Line Brown laying hens in an aviary system

Einfluss der Besatzdichte auf die Legeleistung, Eiqualität und Blutparameter von Hy-Line Brown Legehennen in Volierenhaltung

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Abstract

The effects of stocking density on laying performance, egg quality, leukocyte profile, blood biochemistry, corticosterone levels, litter quality, and noxious gas emission of laying hens in an aviary were investigated. A total of 640 Hy-Line Brown laying hens were kept in an aviary system from 34 to 43 weeks. Birds were randomly assigned to one of four treatments with different stocking density and group size: 13 (pen size = 1.7 × 1.8 × 2.7 m; water nipple = 8; n = 40), 15 (pen size = 1.7 × 1.57 × 2.7 m; water nipple = 7; n = 40), 17 (pen size = 1.7 × 1.38 × 2.7 m; water nipple = 6; n = 40), or 19 (pen size = 1.7 × 1.24 × 2.7 m; water nipple = 5; n = 40) hens per m^2, with each treatment replicated four times. Feed and water were given *ad libitum* for 10 weeks. Hen-day egg production, feed intake, eggshell strength, and egg mass were lower ($P< 0.05$) at 19 birds/m^2 than for other stocking densities. The floor egg rate, heterophil/lymphocyte ratio, and serum corticosterone concentrations were greater ($P< 0.05$) at 19 birds/m^2 than for other stocking densities. Litter moisture and gas emissions (NH_3 and CO_2) were greater ($P< 0.05$) for the 19 birds/m^2 density than for other stocking densities. These results indicate that increasing the density beyond 17 birds/m^2 produces some negative effects on the laying performance of Hy-Line Brown hens kept in this type (Comfort 2 Avairy system, Jansen, The Neatherlands) of aviary system.

Key words

Laying hen, aviary, stocking density, corticosterone, animal welfare, laying performance

Zusammenfassung

Bei Legehennen in Volierenhaltung wurde der Einfluss der Besatzdichte auf Legeleistung, Eiqualität, Leukozytenprofil, Blutbiochemie, Kortikosteronspiegel, Einstreuqualität und schädliche Gasemission untersucht. Insgesamt wurden 640 Hy-Line Brown Legehennen, die in Volieren von der 34. bis zur 43 Woche gehalten wurden, nach dem Zufallsprinzip auf 4 Gruppen mit unterschiedlicher Besatzdichte verteilt: 13 (Stallgröße = 1.7 × 1.8 × 2.7 m; Nippeltränken = 8; n = 40), 15 (Stallgröße = 1.7 × 1.57 × 2.7 m; Nippeltränken = 7; n = 40), 17 (Stallgröße = 1.7 × 1.38 × 2.7 m; Nippeltränken = 6; n = 40), oder 19 (Stallgröße = 1.7 × 1.24 × 2.7 m; Nippeltränken = 5; n = 40) Hennen pro m^2; jede Gruppe mit 4 Wiederholungen. Futter und Wasser wurden über 10 Wochen *ad libitum* verabreicht. Bei einer Besatzdichte von 19 Hennen/m^2 waren die tägliche Eiproduktion je Henne, die Futteraufnahme, die Festigkeit der Eischale und die Eimasse im Vergleich zu allen anderen Versuchsgruppen niedriger ($p < 0.05$). Der Anteil der Bodeneier, das Heterophilen/Lymphozyten Ratio, und die Kortikosteronkonzentration im Serum erreichten bei einer Besatzdichte von 19 Hennen/m^2 höher Werte als in den anderen Gruppen ($p < 0.05$). Die Feuchtigkeit des Einstreus und die Gasemission (NH_3 und CO_2) waren bei einer Besatzdichte von 19 Hennen/m^2 im Vergleich zu den anderen Gruppen ebenfalls erhöht ($p < 0.05$). Die Ergebnisse zeigen, dass sich ein Anstieg der Besatzdichte auf mehr als 17 Hennen/m^2 negativ auf die Legeleistung von Hy-Line Brown Legehennen auswirkt.

Stichworte

Legehenne, Volierenhaltung, Besatzdichte, Kortikosteron, Tierschutz, Legeleistung

Introduction

The welfare of laying hens raised in standard battery cages has been placed under intense scrutiny. Battery cages have been criticised for increasing the incidence of feather damage, foot lesions, and brittle bones (SVEDBERG, 1988, GREGORY and WILKINS, 1989; APPLEBY, 1993; CRAIG and SWANSON, 1994; KANG et al., 2016). Moreover, battery cages restrict the movements of hens and prevent certain behaviours, such as laying eggs in nests, scratching and bathing in sand or soil, and roosting on perches (NICOL, 1987; BAXTER, 1994; ABRAHAMSSON et al., 1996). Although some studies have reported that egg production is similar in aviaries and cages, others have indicated that it is lower in the former, since the risks of feather pecking, cannibalism, disease, and parasites increase (TANAKA and HURNIK, 1992; HANSEN, 1993).

The use of non-cage housing systems for laying hens increased after the 2012 EU ban on battery cages was implemented. European Union Council Directive 1999/74/EC (EUROPEAN COMMUNITIES, 1999) outlined minimum standards for the welfare of laying hens to take effect on January 1, 2012, allotting a maximum stocking density of 9 birds/m^2. However, organisations that began applying the welfare standards on August 3, 1999, were allowed a stocking density of 12 birds/m^2 where the usable area corresponded to the available ground surface (until 2012, when adherence to the new standards was required). The Korean Council Directive 2012–68 (APQA, 2012), which establishes standards for the aviary system for the welfare of laying hens in Korea, imposed a maximum stocking density of 17 birds/m^2. Higher stocking densities have been associated with higher levels of feather pecking in laying hens, in different systems (NICOL et al., 1999). Also laying growers (HANSEN and BRAASTAD, 1994) and growing bantams (SAVORY et al., 1999) showed more feather pecking, measured by the amount of feather damage, under higher stocking densities. Stocking rate has been examined as a factor in a number of epidemiological surveys on the behaviour of laying hens (GUNNARSSON et al., 1999). Numerous studies have demonstrated negative effects of battery cage, floor, or aviary density on egg production, feed intake, rate of floor eggs, and eggshell strength (ROUSH et al., 1984; ANDERSON et al., 1989; ANDERSON et al., 2004; ONBASILAR and AKSOY, 2005; JALAL et al., 2006; KANG et al., 2016). Furthermore, the ratio of heterophils to lymphocytes (H/L), an indicator of stress in birds, increased from 3.59 in control birds to 5.48 in density-stressed birds (KANG et al., 2016). KANG et al. (2016) reported that increased stocking density also increased plasma corticosterone concentrations. In addition, as stocking density increases, the amount of caked litter in the pens increases. The presence of caked litter could serve as a seal on the litter, altering the production of ammonia gas (CARR et al., 1990).

To our knowledge, few studies have examined the physiological effects of stocking density on laying hens in aviary systems. Therefore, this study was conducted to investigate the effects of stocking density of Hy-Line Brown laying hens kept in an aviary system on laying performance, egg quality and blood parameters, as well as noxious gas emission from the litter.

Materials and Methods

The protocol for this experiment was reviewed and approved by the Institutional Animal Care and Welfare Committee of the National Institute of Animal Science, Rural Development Administration, Republic of Korea.

The aviary system

The aviary, a modified Dutch system (Comfort 2 Avairy system, Jansen, The Neatherlands), was divided by wire netting walls into 16 pens. There they were housed in 16 identical pens of an aviary system, including litter (rice hulls), single nests, and three welded wire tiers with nipple drinkers. Nests in multi tiers with automatic egg collection belts were attached to the walls of the room opposite the aviary tiers.

Birds and Experimental design

The experiment included 640 Hy-Line Brown laying hens (*Gallus gallus domesticus*). From 1 d of age, the birds were reared under the same conditions with feed and water supplied on raised slatted floors with high perches. From 5 weeks, they also had access to litter between the slatted areas, At 16 weeks, the pullets were transferred to the laying house. The aviary system was divided by wire netting walls into 16 pens of different size. At 34 weeks of age, the hens were randomly assigned to one of four stocking rates: 13 (pen size = 1.7 × 1.8 × 2.7 m; water nipple = 8; n = 40), 15 (pen size = 1.7 × 1.57 × 2.7 m; water nipple = 7; n = 40), 17 (pen size = 1.7 × 1.38 × 2.7 m; water nipple = 6; n = 40), or 19 (pen size = 1.7 × 1.24 × 2.7 m; water nipple = 5; n = 40) hens per m^2, with each treatment replicated four times. During rearing, the birds were vaccinated against Marek's disease, coccidiosis, and avian encephalomyelitis. Beaks were trimmed at 1 week of age using an automated infrared beak treatment system (NOVA-Tech Engineering Inc., Willmar, Mn).

A commercial-type basal diet was formulated to meet or exceed the nutrient recommendations for laying hens issued by the National Research Council (NRC, 1994) (Table 1). During the 10-week experimental period, hens were provided with feed and water *ad libitum* and were exposed to a 16 h:8 h (light:dark) schedule. The temperature and humidity of the laying house was maintained at 20 ± 3°C and 65%–70% respectively.

Table 1. Composition and nutrient content of experimental diet (as-fed basis)

Zusammensetzung und Nährstoffgehalt der im Versuch eingesetzten Futtermittel

Ingredients, g/kg	
Maize	456.6
Wheat	140.0
Soybean meal	181.7
DDGS	100.0
Soybean oil	9.8
Limestone	99.9
MDCP	6.5
D,L-methionine 99%	1.0
Vitamin premix[1]	1.0
Mineral premix[2]	1.0
Sodium chloride	2.5
Total	1,000.0
Energy and nutrient composition[3]	
ME_n, kcal/kg	2,700.0
Crude protein, g/kg	165.0
Calcium, g/kg	39.0
Available P, g/kg	32.0
Lysine, g/kg	7.9
Methionine + Cystine, g/kg	6.8

[1]Provided per kg of the complete diet: vitamin A (vitamin A acetate), 12,500 IU; vitamin D_3, 2,500 IU; vitamin E (DL-α-tocopheryl acetate), 20 IU; vitamin K_3, 2 mg; vitamin B_1, 2 mg; vitamin B_2, 5 mg; vitamin B_6, 3 mg; vitamin B_{12}, 18 µg; calcium pantothenate, 8 mg; folic acid, 1 mg; biotin, 50 µg; niacin, 24 mg.
[2]Provided per kg of the complete diet: Fe ($FeSO_4 \cdot 7H_2O$), 40 mg; Cu ($CuSO_4 \cdot H_2O$), 8 mg; Zn ($ZnSO_4 \cdot H_2O$), 60 mg; Mn ($MnSO_4 \cdot H_2O$) 90 mg; Mg (MgO) as 1,500 mg.
[3]Nutrient contents in all diet were calculated.

Laying performance

Hen-day egg production rate, rate of floor eggs, broken egg production rate and egg weight were recorded daily, and feed intake and the feed conversion ratio were recorded weekly. Egg mass was calculated as per Hayat et al. (2009).

Egg mass = weekly number of eggs in a replicate × average egg weight

Determination of egg quality parameters

At the end of each week, 10 eggs per replicate were randomly collected. Eggshell strength, eggshell thickness, egg yolk colour, and Haugh units (HU) were measured. Eggshell strength was measured by the Texture Systems Compression Test Cell (model T2100C, Food Technology Co., Ltd., Rockville, MD, USA) and expressed as units of compression force exposed to units of eggshell surface area (kg/cm^2). Eggshell thickness was defined as the mean value of measurements at three different locations on the egg (air cell, equator, and sharp end), measured with a dial pipe gauge (model 7360, Mitutoyo Co. Ltd., Kawasaki, Japan) and calculated using the following formula (Kang et al., 2016):

Eggshell thickness = (sharp point thickness + equator point thickness + air cell thickness)/3.

Egg yolk colour was evaluated by the Roche Yolk Color Fan (Hoffman-La Roche Ltd., Basel, Switzerland; 15 = dark orange; 1 = light pale). Haugh unit values were calculated using a micrometer (model S-8400, Ames, Walthman, MA, USA) with the following formula described by Eisen et al. (1962): $HU = 100 \log (H - 1.7W^{0.37} + 7.6)$, where W is egg weight, and H is albumen height.

Haematological analysis

At the end of the 10-week feeding trial, 2 birds/replicate (i.e., 8 birds per treatment) were selected at random for blood collection. A 5 ml sample of blood was taken from the jugular vein of each bird using EDTA-treated and non-EDTA treated vacutainer tubes (Becton Dicknson, Franklin Lakes, NJ, USA). The whole blood samples were kept on ice and used immediately for haematological analysis. Leukocytes (white blood cells, heterophils, lymphocytes, monocytes, eosinophils, and basophils) from the blood samples were analysed using the Hemavet Multi-Species Hematology System (Drew Scientific Inc., Oxford, CT, USA). The H/L ratio was determined by dividing the number of heterophils by the number of lymphocytes. Serum samples were obtained by centrifuging the samples for 20 min at 25000 *g* at 4°C and were stored at –15°C. Total cholesterol, triglyceride, aspartate aminotransferase (AST), alanine aminotransferase (ALT), and calcium in the serum were quantified using an ADVIA 1650 Chemistry System (Bayer Diagnostic, Puteaux, France). Serum corticosterone concentration was measured using a commercial EIA kit (No. 500655, Caymanm Chemical, Ann Arbor, MI, USA) according to the manufacturer's instructions.

Litter moisture and gas emission

Litter conditions were evaluated at the end of the experiment. Litter quality was assessed in a series of samples that were obtained from four different points located at the edges and in the centre in each compartment. A designated cylindrical sampler, which was 30 cm long and had a diameter of 8 cm, was used to obtain vertical core samples of litter. Each sample was put in a polyethylene bag. Moisture content was measured as described in AOAC method 934.01 (AOAC, 1990). To determine gas emission, litter gas was sampled using a Gastac Gas Sampling Pump (Model GV-100; Gastec Corp., Japan, Gastec Detector Tube No. 3 M and 3 La for ammonia; No. 4LL and 4LK for hydrogen sulphide).

Statistical analysis

All data were analysed by one-way analysis of variance as a completely randomised design using the PROC MIXED procedure (SAS Institute Inc., Cary, NC). Outlier data were identified according to Steel et al. (1997), using the UNIVERIATE procedure of SAS, but no outliers were detected. The pen was an experimental unit for laying performance, egg quality, litter moisture, and gas emission data, whereas the individual bird was the experimental unit for blood parameters, blood biochemistry, and serum corticosterone data. Stocking density was a fixed effect in all statistical models. The LSMEANS procedure was used to calculate mean values. Least squares means were calculated and the means of treatments were compared by the PDIFF option with Tukey adjustment. Significance was set at $P < 0.05$.

Results and Discussion

Laying performance and egg quality

Over the entire experiment (34 to 43 weeks), hen-day egg production, feed intake and egg mass were lower ($P < 0.05$) for the birds in the 19 birds/m^2 treatment than in the other stocking densities (Table 2). The floor egg rate was greater ($P < 0.05$) for 19 birds/m^2 than for other stocking densities. Laying performance declined in response to increased stocking density. This finding supports previous studies that reported decreasing egg production due to a reduction in the amount of feeding area per hen (Hester and Wilson, 1986; Craig and Milliken, 1989; Lee and Moss, 1995; Suto et al., 1997) and increased stocking density (Adams and Craig, 1985).

Table 2. Effects of stocking density on laying performance of laying hens[1]

Einfluss der Besatzdichte auf die Legeleistung von Legehennen

Items	Stocking density (birds/m^2)				SEM[2]	*P*-value
	19	17	15	13		
Hen-day egg production, %	75.6^c	78.9^b	82.3^a	83.1^a	0.90	0.04
Floor eggs, %	4.48^a	2.49^b	2.18^b	2.04^b	0.30	0.03
Broken egg rate,%	2.42	2.17	1.68	2.42	0.40	0.85
Feed intake, g/bird/d	119.9^b	130.8^a	131.8^a	131.1^a	1.18	0.03
Egg mass, g/bird/d	47.2^c	49.5^b	51.2^{ab}	52.1^a	0.64	0.05
Feed conversion ratio	2.54	2.65	2.58	2.52	0.04	0.21

[a,c]Values in a row with no common superscript letter are significantly different ($P< 0.05$).
[1]Data are least squares means of 4 observations per treatment.
[2]Pooled error of mean.

Sarica et al. (2008) and Kang et al. (2016) observed that egg production and egg mass were significantly decreased by higher stocking density. The reduction in growth rate due to stocking density is highly related to a decrease in daily feed intake (Shanawany, 1988). Decreasing egg production has been shown to be attributable to the reduced feeding area per hen and to cannibalism (Lee and Moss, 1995; Sohail et al., 2001; Onbasilar and Aksoy, 2005; Jalal et al., 2006; Mirfendereski and Jahanian, 2015), which is promoted by the stressful conditions of higher stocking density. High stocking densities may also contribute to reduced performance due to the high environmental temperature and the reduced air flow at bird level (Feddes et al., 2002). Metabolic heat dissipation, limited air circulation, and ammonia emissions may all cause performance reduction (Yadgari et al., 2006).

There were no significant differences in egg quality among stocking density treatments (eggshell thickness, eggshell colour, egg yolk colour, and Haugh unit; Table 3). However, eggshell strength was lower ($P< 0.05$) for 19 birds/m^2 than for other stocking density. Similar findings were obtained in other stocking density experiments (Carey et al., 1995; Onbasilar and Aksoy, 2005; Jalal et al., 2006). Guesdon et al. (2006) observed no significant difference in egg weight and shell quality between the layers caged 5 and 6 hens per cage having 660 cm^2 space allowance per hen, but significant increases in broken or cracked eggs were found. Carey et al. (1995) observed that broken-cracked eggs decreased significantly at lower stocking densities. Because few reports are available regarding this topic, more studies may be needed to verify the effects of stocking densitiy on eggshell strength in aviary systems.

Table 3. Effects of stocking density on egg quality of laying hens[1]

Einfluss der Besatzdichte auf die Eiqualität von Legehennen

Items	Stocking density (birds/m^2)				SEM[2]	*P*-value
	19	17	15	13		
Eggshell strength, kg/cm^2	3.92^b	3.96^{ab}	3.97^{ab}	4.07^a	0.23	0.04
Eggshell thickness, µm	378.0	378.6	358.9	366.5	14.25	0.29
Eggshell colour	12.6	12.3	11.7	12.0	0.35	0.85
Egg yolk colour	6.1	5.6	6.0	5.7	0.32	0.23
Haugh unit	95.0	93.5	93.2	94.5	5.25	0.38

[a,b]Values in a row with no common superscript letter are significantly different ($P< 0.05$).
[1]Data are least squares means of 4 observations per treatment.
[2]Pooled error of mean.

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Haematological analysis

There were no significant differences in the level of leukocytes (Table 4). However, the H/L ratio was greater ($P < 0.05$) for 19 birds/m^2 than for other stocking densities. Blood parameters are good indicators of general health status, reflecting physiological, nutritional, and pathological changes; however, it is important to establish appropriate reference values for these parameters (GANONG, 1999; MASOUDI et al., 2011). Leukocyte count has also been used as a measure of immune function in birds (JOHNSON and ZUK, 1998). Many factors, such as exposure to various microbes and chemicals, can cause changes in granulocytic white blood cells (LUCAS and JAMROZ, 1961). The effect of increasing stocking density on blood parameters in poultry requires further research.

Table 4. Effects of stocking density on blood parameters of laying hens[1]

Einfluss der Besatzdichte auf Blutparameter von Legehennen

	Stocking density (birds/m^2)					
Item	19	17	15	13	SEM[2]	*P*-value
Leukocytes[3]						
WBC, K/µl	14.90	14.89	15.35	14.21	2.23	0.82
HE, K/µl	5.22^a	3.52^b	4.28^{ab}	3.58^b	0.37	0.03
LY, K/µl	11.24	9.29	13.74	11.20	1.05	0.32
SI, HE:LY	0.46^a	0.38^b	0.31^b	0.32^b	0.04	0.03
MO, K/µl	1.33	2.11	2.35	1.58	0.23	0.58
EO, K/µl	0.18	0.10	0.12	0.15	0.06	0.17
BA, K/µl	0.03	0.02	0.04	0.02	0.01	0.22

[1]Data are least squares means of 8 observations per treatment.
[2]Pooled error of mean.
[3]Leukocytes: WBC = white blood cells; HE = heterophils; LY = lymphocytes; SI = stress index (H/L-ratio); MO = monocytes; EO = eosinophils; BA = basophils.

The H/L ratio is used by ornithologists as a tool to monitor immune function in birds (DAVIS, 2005). Since GROSS and SIEGEL (1983) first found decreasing numbers of lymphocytes and increasing numbers of heterophils in response to different physiological stressors, the relationship between these leukocytes has become widely accepted as a reliable physiological indicator of stress responses in poultry (MAXWELL, 1993; MAXWELL and ROBERTSON, 1998). The increasing H/L ratio with increasing stocking density indicates increased stress in poultry (MARTRENCHAR et al., 1997; FEDDES et al., 2002; KANG et al., 2016). ONBASILAR et al. (2008) observed that increasing the stocking density significantly increased H/L ratio in broiler chickens. There were no significant differences in serum biochemistry for different stocking rates (Table 5).

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Table 5. Effects of stocking density on blood biochemistry of laying hens[1]

Einfluss der Besatzdichte auf die Blutbiochemie von Legehennen

Item	Stocking density (birds/m^2)				SEM[2]	*P*-value
	19	17	15	13		
Total cholesterol, mg/dl	145.4	106.8	103.7	129.4	14.85	0.25
Triglyceride, mg/dl	1,321.3	1,277.5	1,067.9	1,214.6	181.86	1.25
Glucose, mg/dl	209.2	219.2	217.9	218.4	5.22	0.15
Total protein, mg/dl	5.51	6.38	5.26	5.37	0.29	0.35
Calcium, mg/dl	19.7	23.0	22.7	26.9	3.28	0.42
[3]AST, U/l	212.2	222.7	224.1	206.1	7.06	0.11
[4]ALT, U/l	2.14	3.01	2.25	2.04	0.34	0.23

[1]Data are least squares means of 8 observations per treatment.
[2]Pooled error of mean.
[3]AST = Aspartate aminotransferase.
[4]ALT = Alanine aminotransferase.

Serum corticosterone

Serum corticosterone was greater ($P < 0.05$) for 19 birds/m^2 than for other stocking densities (Figure 1). The corticosterone concentration level was 757.0, 461.4, 393.4, and 337.4 pg/ml for 19, 17, 15, and 13 birds/m^2, respectively. Serum corticosterone concentration has been widely used as a measure of environmental stress in poultry (PESTI and HOWARTH, 1983; CRAIG and VARGAS, 1986; MCFARLANE and CURTIS, 1989; ZULKIFLI et al., 2003). CHENG and MUIR (2004) reported that laying hens showed significantly lower plasma corticosterone levels in single-bird cages (0.53 m^2/bird) than in 10-bird cages (0.42 m^2/bird), indicating that social stressors could induce higher production of corticosterone in hens. Later work has showed that serum corticosterone levels increase when population density increase, presumably because birds are forced to compete for feeding and watering space (TURKYILMAZ, 2008; KANG et al., 2016).

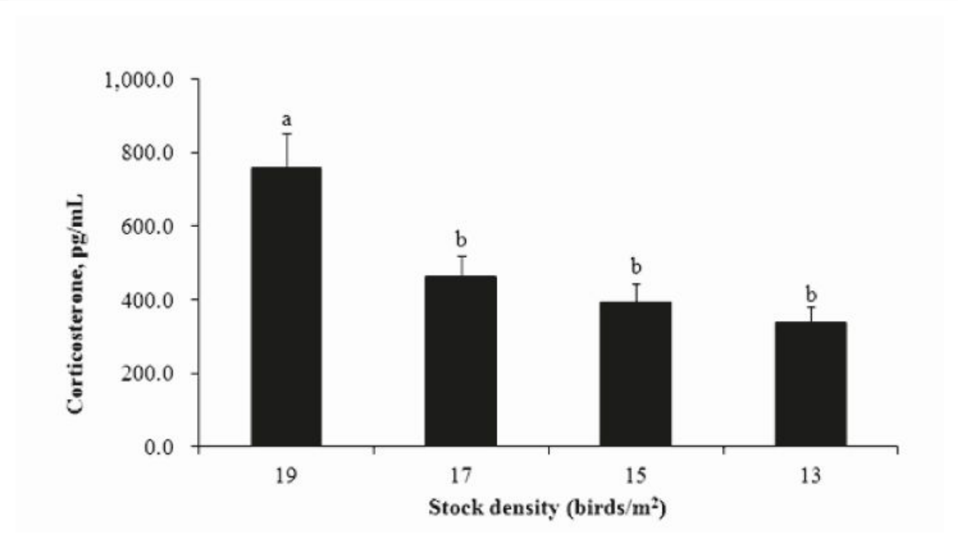

Figure 1. Effects of stocking density on serum corticosterone concentration of laying hens[1]

[a,b]Values in row with no common superscripts letter are significantly different (*P*< 0.05).

[1]Data are least squares means of 8 observations per treatment.

Einfluss der Besatzdichte auf die Kortikosteronkonzentration im Serum von Legehennen[1]

[a,b] Werte mit unterschiedlichen Buchstaben unterscheiden sich signifikant (*P*< 0.05).

[1]Kleinste mittlere Quadrate von 8 Beobachtungen je Versuchsgruppe

Litter moisture and gas emissions

Litter moisture levels and gas emissions (CO_2 and NH_3) were greater (P< 0.05) for 19 birds/m^2 than for the 17, 15, and 13 birds/m^2 stocking densities (Table 6). This effect may be due to the increase in litter moisture with the higher stocking density (HERMANS and MORGAN, 2007). As stocking density increased, the amount of caked litter in the pens also increased. Higher stocking density increases nitrogen and moisture level in the litter and thus favours microbial activity. The caked litter could have served as a seal, altering the production of ammonia. Also, caked litter corresponds to high litter moisture or areas where litter becomes anaerobic, which suppresses ammonia volatilisation (CARR et al., 1990). KRISTENSEN and WATHES (2000) reported that ammonia is formed in litter by bacterial breakdown of litter and that ammonia concentrations are increased by moisture, high temperature, and poor ventilation. AL-HOMIDIAN and ROBERTSON (2003) reported that increases in stocking density resulted in increased concentration of ammonia. The wetter the litter, the more likely it will promote the proliferation of pathogenic bacteria and moulds. Wet litter is also the primary causes of ammonia emission, one of the most serious performance and environmental factors affecting poultry production (STEPHEN, 2012). Litter condition is governed by the type of material, death, friability, and moisture as well as housing, technical equipment, and management. From a welfare point a view, a high stocking density may create various problems such as increased ammonia and heat produced from the birds, which can lead to stressful conditions and cause the death of individual hens (KANG et al., 2016).

Table 6. Effects of stocking density on litter moisture and gas emission of litter

Einfluss der Besatzdichte auf die Feuchtigkeit des Einstreus und der Gasemission von Legehennen

	Stocking density (birds/m^2)					
Item	19	17	15	13	SEM[2]	*P*-value
Litter moisture, %	40.93[a]	22.93[b]	23.57[b]	23.67[b]	1.39	0.04
Gas emission						
NH_3, ppm	9.07[a]	5.70[b]	5.85[b]	5.63[b]	0.32	0.03
CO_2, ppm	755.6[a]	663.9[b]	611.8[b]	611.3[b]	24.25	< 0.01

[a,b]Values in a row with no common superscript letter are significantly different ($P < 0.05$).
[1]Data are least squares means of 4 observations per treatment.
[2]Pooled error of mean.

Conclusions

In conclusion, increasing stocking density over 17 hens per m^2 (i.e., 19 per m^2 hens) negatively affected the laying performance, litter quality, and noxious gas emission (NH_3 and CO_2). However, for the purpose of defining a general optimal stocking density, it is necessary to consider laying hen genotype and environmental conditions. As stocking density directly influences egg production economy, there is a need to find compromises, considering suggestions in the literature of top limits at the level of 13 to 17 per m^2.

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References

ABRAHAMSSON, P., R. TAUSON, M.C. APPLEBY, 1996: Behaviour, health and integument of four hybrids of laying hens in modified and conventional cages. Br. Poult. Sci. **37,** 521-540.

ADAMS, A.W., J.V. CRAIG, 1985: Effect of crowding and cage shape on productivity and profitability of caged layers: a survey. Poult. Sci. **64,** 238-242.

AL-HOMIDIAN, A., J.F. ROBERTSON, 2003: Effect of litter type and stocking density on ammonia dust concentrations and broiler performance. Br. Poult. Sci. **44,** S7-S8.

ANDERSON, K.E., A.W. ADAMS, J.V. CRAIG, 1989: Behavioral adaptation on floor-reared white leghorn pullets to different cafedensities and cage shapes during the initial settling in period. Poult. Sci. **68,** 70-78.

ANDERSON, K.E., G.S. DAVIS, P.K. JENKINS, A.S. CARROLL, 2004: Effect of bird age, density and molt on behavioral profiles of two commercial layer strains in cages. Poult. Sci. **83,** 15-23.

AOAC, 1990: Official Method of Analysis. 15th ed. Association of Official Analytical Chemists, Washington, DC.

APPLEBY, M.C., 1993: Should cages for laying hens be banned or modified? Animal Welfare, **2,** 67-80.

APQA, 2012: Livestock farm animal welfare certification standards (regulation No. 2012-68). Animal and Plant Quarantine Agency, Republic of Korea.

BAXTER, M.R., 1994: The welfare problems of laying hens in battery cages. Vet. Rec. **134,** 614-619.

CAREY, J.B., F.L. KUO, K.E. ANDERSON, 1995: Effect of cage population on the productive performance of layers. Poult. Sci. **74,** 633-637.

CARR, L.E., F.W. WHEATON, L.W. DOUGLASS, 1990: Empirical models to determine ammonia concentrations from broiler chicken litter. Transactions of the ASAE **33,** 1337-1342.

CHENG, H.W., W.M. MUIR, 2004: Chronic social stress differentially regulates neuroendocrine response in laying hens: II. Genetic basis of adrenal responses under three different social conditions. Psychoneuroendocrinlogy **29,** 961-971.

CRAIG, J.V., J.C. SWANSON, 1994: Review: welfare perspectives on hens kept for egg production. Poult. Sci. 73, 921-938.

CRAIG, J.V., J. VARGAS, 1986: Corticosteroids and other indicators of hens' well-being in four laying house environments. Poult. Sci. **65,** 856-863.

CRAIG, J.V., G.A. MILLIKEN, 1989: Further studies of space and group size effect in caged hens of stock differing in fearful behavior. Productivity and behavior. Poult. Sci. **68,** 9-16.

DAVIS, A.K., 2005: Effect of handling time and repeated sampling on avian white blood cell counts. J. Field Ornithology, **76,** 334-338.

GANONG, W.F., 1999: Review of medicinal physiology. 19th ed. Stanford, Connecticut, Appleton an Lange, P. 353.

GREGORY, N.G., L.J. WILKINS, 1989: Broken bones in domestic fowl: Handling and processing damage in end-of-lay battery hens. Br. Poult. Sci. **30,** 555-562.

GROSS, W.B., H.S. SIEGEL, 1983: Evaluation of the heterophil/lymphocyte ratio as a measure of stress in chickens. Avian Diseases, **27,** 972-979.

GUESDON, V., A.M.H. AHMED, S. MALLET, J.M. FAURE, Y. NYS, 2006: Effect of beak trimming and cage design on laying hen performance and egg quality. Br. Poult. Sci. **47,** 1-12.

GUNNARSSON, S., L.J. KEELING, J. SVEDBERG, 1999: Effect of rearing factors on the prevalence of floor eggs, cloacal cannibalism and feather pecking in commercial flocks of loose housed laying hens. Br. Poult. Sci. **40,** 12-18.

EISEN, E.J., B.B. BOHREN, H.E. MCKEAN, 1962: Haugh unit as a measure of egg albumen quality. Poult. Sci. **41,** 1361-1368.

EUROPEAN COMMUNITIES, 1999: Council Directive 1999/74/EC/. Official Journal of the European Communities, L203/53.

FEDDES, J.J.R., E.J. EMMANUEL, M.J. ZUIDHOF, 2002: Broiler performance, body weight variance, feed and water intake, and carcass quality at different stocking densities. Poult. Sci. **81,** 774-779.

HANSEN, I., 1993: Ethological studies of laying hens in aviaries and cages. Ph. D. Thesis, Agricultural University of Norway, As Norway.

HANSEN, I., B.O. BRAASTAD, 1994: Effect of rearing density on pecking behavior and plumage condition of laying hens in 2 types of aviary. Appl. Anim. Behav. Sci. **40,** 263-272.

HAYAT, Z., G. GHERIAN, T.N. PASHA, F.M. KHATTAK, M.A. JABBER, 2009: Effect of feeding flax and two types of antioxidants on egg production, egg quality, and lipid composition eggs. J. Appl. Poult. Res. **18,** 541-551.

HESTER, P.Y., E.K. WILSON, 1986: Performance of White leghorn hens in response to cage density and the introduction of cage mates. Poult. Sci. **65,** 2029-2033.

HERMANS, P.G., K.L. MORGAN, 2007: Prevalence and associated risk factors of necrotic enteritis on broiler farms in the United Kingdom; A cross-sectional survey. Avian Pathol. **36,** 43-51.

JALAL, M.A., S.E. SCHEIDELSER, D. MARX, 2006: Effect of bird cage space and dietary metabolizable energy level on production parameters in laying hens. Poult. Sci. **85,** 306-311.

JOHNSON, T.S., M. ZUK, 1998: Parasites, morphology, and blood characters in male red jungle fowl during development. Condor, **100,** 749-752.

KANG, H.K., S.B. PARK, S.H. KIM, C.H. KIM, 2016: Effects of stock density on the laying performance, blood parameter, corticosterone, litter quality, gas emission and bone mineral density of laying hens in floor pens. Poult. Sci. **95,** 2764-2770.

KRISTENSEN, H.H., C.M. WATHES, 2000: Ammonia and poultry welfare: a review. World's Poult. Sci. J. **56,** 235-245.

LEE, K., C.V. MOSS, 1995: Effect of population density on layer performance. Poult. Sci. **74,** 1754-1760.

LUCAS, A.M., C. JAMROZ, 1961: Atlas of Avian Hematology. USDA Monograph 25. USDA, Washongton DC.

MARTRENCHAR, A., J.P. MORISSE, D. HUONNIC, J.P. COTTE, 1997: Influence of stock density on some behavioural, physiological, and productivity traits of broilers. Vet. Res. **28,** 473-480.

MASOUDI, A., M. CHAGI, M. BOJAROPOUR, K. MIRZADEH, 2011: Effects of different levels of date pits on performance, carcass characteristics and blood parameters of broiler chickens. J. Appl. Poult. Res. **39,** 399-405.

MAXWELL, M.H., 1993: Avian blood leucocyte responses to stress. World's Poult. Sci. J. **49,** 34-43.

MAXWELL, M.H., G.W. ROBERTSON, 1998: The avian heterophil leucocyte: a review. World's Poult. Sci. J. **54,** 155-178.

MCFARLANE, J.M., S.E. CURTIS, 1989: Multiple concurrent stressors in chicks. 3. Effects on plasma corticosterone and the heterophil:lymphocyte ratio. Poult. Sci. **68,** 522-527.

MIRFENDERESKI, E., R. JAHANIAN, 2015: Effects of dietary organic chromium and vitamin C supplementation on performance, immune response, blood metabolites, and stress status of laying hens subjected to high stocking density. Poult. Sci. **94**, 281-288.

NICOL, C.J., 1987 Behavioural responses of laying hens following a period of spatial restriction. Animal Behav. **35,** 1709-1719:.

NICOL, C.J., N.G. GREGORY, T.G. KNOWLES, I.D. PARKMAN, L.J. WILKINS, 1999: Differential effects of increased stocking density, mediated by increased flock size, on feather pecking and aggression in laying hens. Appl. Anim. Behav. Sci. **65,** 137-152.

NRC, 1994: Nutrient Requirements of Poultry, 9th Rev edn. National Acadaemy Press, Washington, DC.

ONBASILAR, E.E., F.T. AKSOY, 2005: Stress parameters and immune response of layers under different cage floor and density conditions. Livest. Prod. Sci. **95,** 255-263.

ONBASILAR, E.E., O. POYRAZ, S. CETIN, 2008: Effects of breeder age and stocking density on performance, carcass characteristics and some stress parameters of broilers. Asian-Aust. J. Anim. Sci. **21,** 262-269.

PESTI, G.M., B. HOWARTH, 1983: Effects of population density on the growth, organ weights, and plasma corticosterone of young broiler chicks. Poult. Sci. **62,** 1080-1083.

ROUSH, W.B., M.M. MASHALY, H.B. GRAVES, 1984: Effect of increased bird production in a fixed cage area on production and economic response of Single Comb White leghorn laying hens. Poult. Sci. **63,** 45-48.

SARICA, M.S., S. BOGA, U.S. YAMAK, 2008: The effects of space allowance on egg yield, egg quality and plumage condition of laying hens in battery cages. Czech J. Anim. Sci. **53,** 346-353.

SAVORY, C.J., J.S. MANN, M.G. MACLEOD, 1999: Incidence of pecking damage in growing bantams in relation to food form, group size, stocking density, dietary tryptophan concentration and dietary protein source. Br. Poult. Sci. **40,** 579-584.

SHANAWANY, M.M., 1988: Broiler performance under high stocking densities. Br. Poult. Sci. **29,** 43-52.

SOHAIL, S.S., M.M. BRYANT, S.K. RAO, D.A. ROLAND, 2001: Influence of cage space and prior dietary phosphorus level on phosphorus requirement of commercial leghorns. Poult. Sci. **80,** 769-775.

STEEL, R.G.D., J.H. TORRIE, D.A. DICKEY, 1997: 1997 Principles and procedures of Statistics: A Biochemical Approach 3rd ed. McGraw Hill Book Co., New York, NY.

STEPHEN, R.C., 2012: Nutrition and wet litter problems in poultry. Anim. Feed Sci. & Technol. **173,** 65-75.

SUTO, Z., P. HORN, J. UJVARI, 1997: The effect of different housing systems on production and egg quality traits of brown and leghorn type layers. Acta Agraria Kaposvariensis, **1,** 29-35.

SVEDBERG, J., 1988: The connection between environment and foot condition in laying hens. 6th International Congress on Animal Hygiene, Skara. 125-130.

TANAKA, T., J.F. HURNIK, 1992: Comparison of behaviour and performance of laying hens housed in battery cages and an aviary. Poult. Sci. **71,** 235-243.

TURKYILMAZ, M.K., 2008: The effect of stocking density on stress reaction in broiler chickens during summer. Turkish J. Vet. & Anim. **32,** 31-36.

YADGARI, L., R. KINEREICH, S. DRUYAN, A. CAHANER, 2006: The effects of stocking density under hot conditions on growth, meat yield and meat quality of featherless and feathered broilers. World’s Poult. Sci. J. **62,** 603-604.

ZULKIFLI, I., P.K. LIEW, D.A. ISRAF, A.R. OMAR, M. HAIR-BEJO, 2003: Effect of early age feed restriction and heat conditioning on heterophil/lymphocyte ratios, heat shock protein 70 expression and body temperature of heat-stressed broiler chickens. J. Therm. Biol. **28,** 217-222.